Existing Solution	What it does well	Limitation	How SignalLens is different
Waze / Google Maps	Real-time crowd reports	Closed system, no evidence trail, no open contribution	Open-source + full evidence provenance
StormPath, CanaanRoads, icemap	Community reporting + maps	Little or no AI classification	Adds zero-shot AI + weather context + transparent confidence
CivicTrack / FixMyStreet style apps	Citizen complaint tracking	Mostly manual categories	Automatic zero-shot classification + cross-modal evidence
Official weather / disaster apps	Authoritative alerts	Usually national-level, slow, no local photo evidence	Hyper-local + user reports + AI triage
Pure social media	Fast information	Noisy, unverified, hard to filter	Structured evidence panel + freshness + status

1. Problem Statement

Local disruption information is fragmented and hard to trust.

When heavy rain, waterlogging, road closures, fallen trees, or infrastructure damage occur, useful information is scattered across:

- Weather apps
- Government notices
- WhatsApp / social media messages
- Informal community reports

People (students, commuters, campus staff, residents) have no single place to quickly understand:

- What is being reported?
- How fresh is the information?
- What evidence supports it?
- Is it verified or just a rumour?

Existing tools either collect reports without intelligence, or use AI without showing clear evidence and limitations.

2. What Problem Does SignalLens Solve?

SignalLens turns scattered reports + optional photos + live weather data into a **clear, evidence-backed view** of local disruptions.

It answers four practical questions:

1. **What happened?** → Concise AI-categorised summary
2. **What evidence supports it?** → Original text, timestamp, source, optional photo analysis, weather context
3. **How confident should I be?** → Explicit “Unverified / Needs Review” status + model scores shown as predictions, not facts
4. **What should I do next?** → Cautious, practical suggestion based on available evidence

It is designed as an **information-assistance tool**, never as an official emergency system.